



ABLATION RUN - batch\_size=32  
Random loss baseline:  $\log(32) = 3.466$



```
Downloading: "https://download.pytorch.org/models/resnet50-0676ba61.pth" to /root/.cache/torch/hub/checkpoints/resnet50-0676ba61.pth
```

100% | 97.8M/97.8M [00:00<00:00, 163MB/s]

```
model.safetensors: 100% ██████████ 268M/268M [00:02<00:00, 120MB/s]
```

|             |              |                   |
|-------------|--------------|-------------------|
| Epoch 01/08 | train 1.7631 | val 1.3874 ← best |
| Epoch 02/08 | train 1.0380 | val 1.2473 ← best |
| Epoch 03/08 | train 0.7031 | val 1.2639        |
| Epoch 04/08 | train 0.4734 | val 1.1973 ← best |
| Epoch 05/08 | train 0.3061 | val 1.2211        |
| Epoch 06/08 | train 0.2131 | val 1.1923 ← best |
| Epoch 07/08 | train 0.1490 | val 1.1798 ← best |

```
/content/mini-clip-DLAI/src/train.py:136: FutureWarning: You are using `torch.load` with `weights_only=False` (the current default value), which uses the default pickle module (which is insecure). To silence this warning, use `torch.load` with `weights_only=True` (which is the recommended option). To learn more about this warning, see the torch.load documentation.
  checkpoint = torch.load(load_path, map_location=device)
```

Epoch 08/08 | train 0.1214 | val 1.1869

batch\_size=32

| Metric       | Score  |
|--------------|--------|
| Text → Image |        |
| R@1          | 20.00% |
| R@5          | 45.80% |
| R@10         | 59.30% |
| Image → Text |        |
| R@1          | 18.60% |
| R@5          | 46.60% |
| R@10         | 61.50% |

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ABLATION RUN - batch\_size=64

Random loss baseline:  $\log(64) = 4.159$

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|             |              |                   |
|-------------|--------------|-------------------|
| Epoch 01/08 | train 2.2580 | val 1.8177 ← best |
| Epoch 02/08 | train 1.3108 | val 1.5859 ← best |
| Epoch 03/08 | train 0.8693 | val 1.5420 ← best |
| Epoch 04/08 | train 0.5797 | val 1.5356 ← best |
| Epoch 05/08 | train 0.3723 | val 1.4874 ← best |
| Epoch 06/08 | train 0.2559 | val 1.4792 ← best |
| Epoch 07/08 | train 0.1838 | val 1.4761 ← best |
| Epoch 08/08 | train 0.1520 | val 1.4830        |

batch\_size=64

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Metric

Score

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Text → Image

|      |        |             |
|------|--------|-------------|
| R@1  | 21.10% | <div></div> |
| R@5  | 48.20% | <div></div> |
| R@10 | 61.70% | <div></div> |

Image → Text

|      |        |             |
|------|--------|-------------|
| R@1  | 21.60% | <div></div> |
| R@5  | 50.00% | <div></div> |
| R@10 | 62.80% | <div></div> |

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ABLATION RUN - batch\_size=128

Random loss baseline:  $\log(128) = 4.852$

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Epoch 01/08 | train 3.0937 | val 2.4340 ← best  
Epoch 02/08 | train 1.8495 | val 2.0583 ← best  
Epoch 03/08 | train 1.2456 | val 1.9938 ← best  
Epoch 04/08 | train 0.8312 | val 2.0185  
Epoch 05/08 | train 0.5723 | val 1.9239 ← best  
Epoch 06/08 | train 0.4175 | val 1.9292  
Epoch 07/08 | train 0.3057 | val 1.8755 ← best  
Epoch 08/08 | train 0.2733 | val 1.8598 ← best

batch\_size=128

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Metric

Score

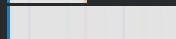
---

Text → Image

R@1 21.80%



R@5 46.90%



R@10 62.80%



Image → Text

R@1 21.20%



R@5 49.80%



R@10 62.30%



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All ablation runs complete.